IUPAC naming

Settings

PRACTICING Alk

Problem 1

MOLECULE

IUPAC NAMING · QUICK START

Choose what to practice

We've picked what most courses cover first. Adjust it to match yours, or just start.

Carbon skeleton

Pick at least one

- [x] Alkane
- [x] Alkene
- [x] Alkyne
- [ ] Aromatic

Functional groups

Leave all unchecked for plain hydrocarbons

- [x] Alcohol
- [x] Alkyl halide
- [x] Ether
- [ ] Thiol
- [ ] Aldehyde
- [ ] Ketone
- [ ] Epoxide

> **More functional groups** Carboxylic acid derivatives, amine, nitro

> **Size, rings and stereochemistry** 3–10 carbons · rings on · stereochemistry off

> **Advanced options** 2°/3° counts, ring topology, aromatic positions, group count and distance, custom fragments

Saved to Settings › Molecules. Change it any time.

Start practicing →

Name this mole

Type the I

Skip

Next problem →

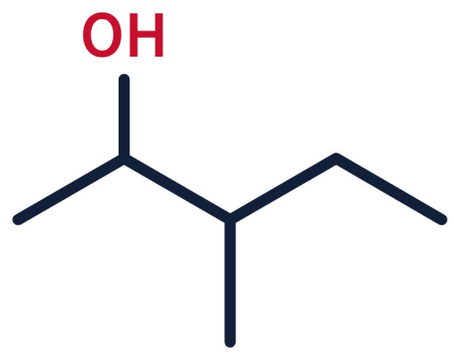
IUPAC naming

Settings

PRACTICING Alkane, alkene, alkyne | Alcohol, alkyl halide, ether

Problem 7

MOLECULE



Name this molecule

Type the IUPAC name

Submit | Show answer | Skip

Feedback appears here after Submit or Show answer

Next problem →

This session 5 of 6 correct **83%** 1 skipped · 0 answers shown

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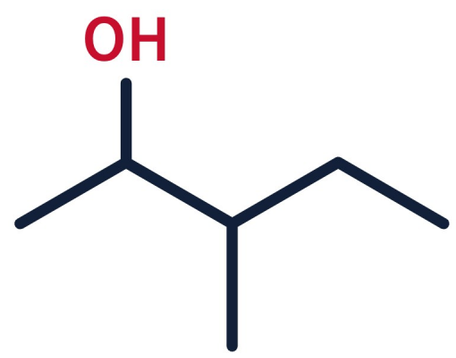
IUPAC naming

Settings

PRACTICING Alkane, alkene, alkyne | Alcohol, alkyl halide, ether

Problem 7

MOLECULE



Name this molecule

3-methyl-2-pentanol

Submit | Show answer | Skip

Correct

Your answer 3-methyl-2-pentanol

Preferred name **3-methylpentan-2-ol**

Synonyms 3-methyl-2-pentanol (your answer)

Press Enter for the next problem

Next problem →

This session 6 of 7 correct **86%** 1 skipped · 0 answers shown

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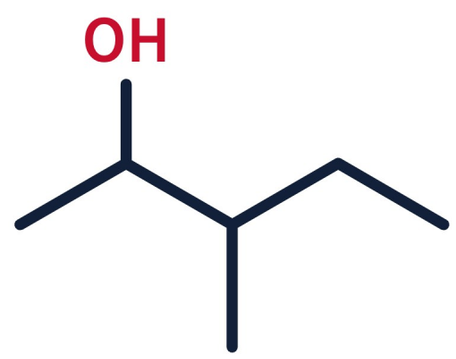
IUPAC naming

Settings

PRACTICING Alkane, alkene, alkyne · Alcohol, alkyl halide, ether

Problem 7

MOLECULE



Name this molecule

3-methylpentan-4-ol

Submit · Show answer · Skip

Not quite

Look at the number in front of **-ol**. Which end of the chain should you number from so the OH group gets the lowest number?

Press Enter to submit again

Next problem →

This session 5 of 7 correct **71%** 1 skipped · 0 answers shown

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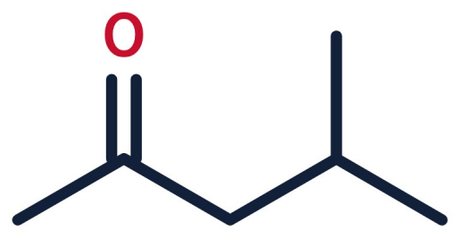
IUPAC naming

Settings

PRACTICING Alkane, alkene, alkyne | Alcohol, alkyl halide, ether

Problem 9

MOLECULE



Name this molecule

4-methylpentan-

Submit | Show answer | Skip

Answer shown

Correct name **4-methylpentan-2-one**

Synonyms 4-methyl-2-pentanone
methyl isobutyl ketone
isobutyl methyl ketone

Your attempt 4-methylpentan- (not graded)

Press Enter for the next problem

Next problem →

This session 6 of 8 correct **75%** 1 skipped · 1 answer shown

Unlimited practice for the organic chemistry problems that trip students up on exams.

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ALL GENERATORS

Molecules

GENERATORS

IUPAC naming

Synthesis

Acid/Base Coming soon

IUPAC naming

Settings that only apply to the IUPAC naming generator.

MOLECULE OPTIONS

Use the shared Molecules settings
Semester 1 · 7 groups · 3–10 carbons · rings on · stereo off · Edit shared

Customize for IUPAC naming only
Changes here won't affect other generators.

NAMING PROBLEMS

Parent chain length
Ring size for aromatic parents. 3 to 10

Most senior functional group
Chosen from the groups allowed under Molecule options. Any allowed

Number of principal groups 1 to 2

Number of secondary groups 0 to 2

PRACTICE

Count skipped problems in my score

Keyboard shortcuts
Enter submits · Enter again for next · Esc skips

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Quick start

Choose what to practice

We've picked what most courses cover first. Adjust it to match yours, or just start.

Carbon skeleton

Pick at least one

- [x] Alkane 
- [x] Alkene 
- [x] Alkyne 
- [ ] Aromatic 

Functional groups

Leave all unchecked for plain hydrocarbons

- [x] Alcohol 
- [x] Alkyl halide 
- [x] Ether 
- [ ] Thiol 
- [ ] Aldehyde 
- [ ] Ketone 
- [ ] Epoxide 

> More functional groups

Carboxylic acid derivatives, amine, nitro

> Size, rings and stereochemistry

3–10 carbons · rings on · stereochemistry off

> Advanced options

2°/3° counts, ring topology, aromatic positions, group count and distance, custom fragments

Start practicing →

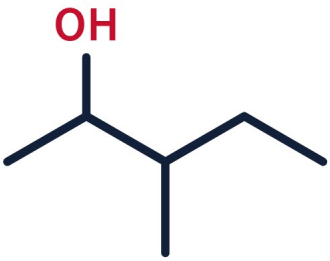
Saved to Settings › Molecules. Change it any time.

IUPAC naming

Alkane, alkene, alkyne | Alcohol, alkyl halide, ether

Problem 7 **5/6 · 83%**

MOLECULE



Name this molecule

Type the IUPAC name

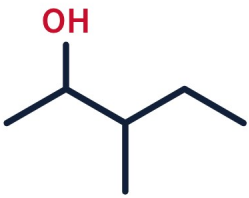
Submit

Show answer | Skip

IUPAC naming

Problem 7 5/6 · 83%

MOLECULE



Name this molecule

3-methylpentan-2-

Submit | Show answer | Skip

Suggestions off

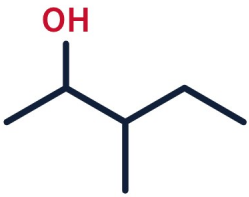
Submit

IUPAC naming

Alkane, alkene, alkyne | Alcohol, alkyl halide, ether

Problem 7 **6/7 · 86%**

MOLECULE



Name this molecule

3-methyl-2-pentanol

Correct

Preferred name

3-methylpentan-2-ol

Synonyms

3-methyl-2-pentanol (your answer)

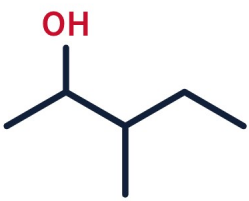
Next problem →

IUPAC naming

Alkane, alkene, alkyne | Alcohol, alkyl halide, ether

Problem 7 **5/7 · 71%**

MOLECULE



Name this molecule

3-methylpentan-4-ol

Not quite

Look at the number in front of **-ol**. Which end of the chain should you number from so the OH group gets the lowest number?

Submit again

Show answer | Skip

✓ Your choices are saved in Settings › Molecules. Open Settings Got it

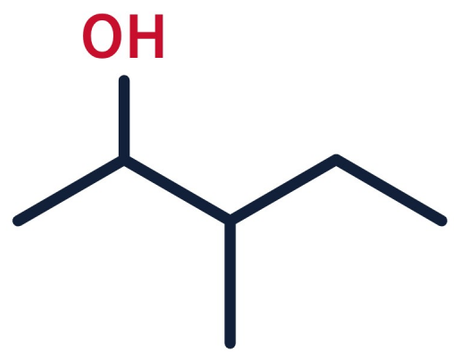
IUPAC naming

Settings

PRACTICING Alkane, alkene, alkyne | Alcohol, alkyl halide, ether

Problem 1

MOLECULE



Name this molecule

Type the IUPAC name

Submit | Show answer | Skip

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Next problem →

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- [x] Ether
- [ ] Thiol
- [ ] Aldehyde
- [ ] Ketone
- [ ] Epoxide

More functional groups

Carboxylic acid derivatives, amine, nitro

Carboxylic acid derivatives

- [ ] Carboxylic acid
- [ ] Acid chloride
- [ ] Anhydride
- [ ] Ester
- [ ] Amide
- [ ] Nitrile

Auxiliary

- [ ] Amine
- [ ] Nitro

Size, rings and stereochemistry

3–10 carbons · rings on · stereochemistry off

Molecule size
Number of carbons in each molecule.
3 to 10

Include rings
Cycloalkanes and other cyclic molecules.
On

Include heteroatoms in rings
Pyridine, furan, thiophene and similar. Needs Aromatic checked.
Off

Stereochemistry
When included, answers must give correct R/S and E/Z.
Off | Include

Advanced options

2°/3° counts, ring topology, aromatic positions, group count and distance, custom fragments

	CARBONS		AMINES (needs Amine checked)	
Secondary (2°) Carbon bonded to 2 carbons · R_2NH	0 to	4	0 to	1
Tertiary (3°) Carbon bonded to 3 carbons · R_3N	0 to	2	0 to	1

Alkyl substituents (R)
Which alkyl substituents can branch off the parent chain or ring.
Select all · Clear

1 carbon

- [x] Methyl — methyl

2 carbons

- [x] Ethyl — ethyl

3 carbons

- [x] Propyl — propyl
- [x] Isopropyl — propan-2-yl

4 carbons

- [x] Butyl — butyl
- [x] Isobutyl — 2-methylpropyl
- [x] sec-Butyl — butan-2-yl
- [x] tert-Butyl — 2-methylpropan-2-yl

5 carbons

- [x] Pentyl — pentyl
- [x] Isopentyl — 3-methylbutyl
- [ ] Neopentyl — 2,2-dimethylpropyl
- [x] tert-Pentyl — 2-methylbutan-2-yl
- [ ] sec-Pentyl — pentan-2-yl
- [ ] 3-Pentyl — pentan-3-yl

- [x] **Allow more complicated alkyl groups**
  Longer or more highly branched side chains than the ones above, such as 2-ethylbutyl or 3,3-dimethylbutyl.

  Functional groups in these side chains

  - (•) **Not allowed** — Side chains are carbon and hydrogen only.
  - ( ) **Allowed** — Side chains can carry the functional groups checked in Molecules, e.g. 2-hydroxyethyl or 3-chloropropyl.

Ring topology
Single rings are always included when Include rings is on.
- [ ] Spiro
- [ ] Fused
- [ ] Bridged

Aromatic ring positions
Substituent patterns on disubstituted benzene rings. Needs Aromatic checked.
- [ ] Ortho (1,2)
- [ ] Meta (1,3)
- [ ] Para (1,4)

Functional group count
Total number of functional groups in each molecule.
0 to 3

Functional group distance
How far apart two functional groups are (Warren, *The Disconnection Approach*).
- [x] 1,1
- [x] 1,2
- [x] 1,3
- [x] 1,4
- [x] 1,5

Custom fragments
Every generated molecule must contain all of these fragments.
No fragments added.
+ Add fragment

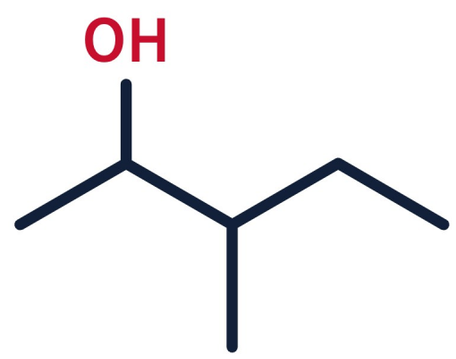
IUPAC naming

Settings

PRACTICING Alkane, alkene, alkyne · Alcohol, alkyl halide, ether

Problem 7

MOLECULE



Name this molecule

3-methylpentan-4-ol

Submit · Show answer · Skip

Not quite

Generating feedback…

Press Enter to submit again

Next problem →

This session 5 of 7 correct **71%** 1 skipped · 0 answers shown

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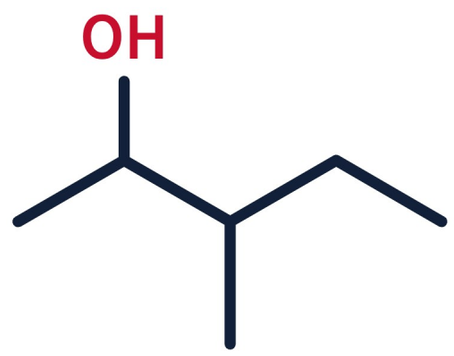
IUPAC naming

Settings

PRACTICING Alkane, alkene, alkyne | Alcohol, alkyl halide, ether

Problem 7

MOLECULE



Name this molecule

3-methylpentan-2-ol

Submit | Show answer | Skip

Correct

Preferred name **3-methylpentan-2-ol**

On your second try. Only first tries count toward your score.

Press Enter for the next problem

Next problem →

This session 5 of 7 correct **71%** 1 skipped · 0 answers shown

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